

Exercise Sheet: Cybersecurity Fundamentals

Exercise 1

1.1: Explain, in your own words, the three protection goals that make up the C-I-A triad. Which two protection goal were added later on?

1.2: How was risk mathematically defined in the lecture?

1.3: Which questions are indicative of a security mindset?

1.4: What are the three steps when creating an attack tree?

1.5: What is the defining property of symmetric encryption opposed to asymmetric encryption?

1.6: What are the three defining properties of cryptographic hash functions?

1.7: Briefly explain, in your own words, the basic idea behind hybrid encryption.

Exercise 2

Assume your goal is to get hold of the written exam for a specific course. You know that the laptop of the respective lecturer is consistently unlocked when they go to the bathroom. However, their office is on a restricted corridor.

2.1: Draw an attack tree for this scenario. Consider all three steps of creating an attack tree separately and draw the tree in all three steps. You don't need a comprehensive tree, but make sure to have at least a few full paths with countermeasures.

2.2: How does the tree change, if the lecturer shares the office with another lecturer? Draw the tree again or add a respective sub-tree in this scenario.

Exercise 3

3.1: Explain the term Layered Security in your own words.

Exercise 4

Assume a scenario like the "red telephone", where two parties need to repeatedly communicate confidential information.

4.1: Which type of encryption is suitable for this scenario? Explain your answer.

4.2: Assume the number of parties grows to five. Which type of encryption is suitable for this scenario? Explain your answer.

Exercise 5

5.1: What are the benefits of hybrid encryption?

5.2: Can you imagine a scenario where this is used today? Give an example and explain why it makes sense in this case.

Exercise 6

6.1: Encrypt the bitstring $m = 10110110\ 01100111$ using with the key $k = 01010110\ 01011100$ using the One-Time-Pad algorithm. Assume bit-wise encryption.